SOP – PRODUCT DEVELOPMENT LIFECYCLE

[CONFIDENTIAL]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1.	0.1 draft	New product development process defined	No process in place	Anitha N 10-Nov-23	Santosh V 10-11-23	Santosh V 10-11-23
2.	Issue B Version 1.0		All review comments fixed	Anitha N 10-Nov-23	Santosh V 20-11-23	Santosh V 21-11-23
2.	Issue B Version 1.1	Section 9.9 Addition of product retirement process	Process enhancement	Sachin Aramani 13-Mar-24	Anitha N 12-Apr-24	Santosh V 12-Apr-24
3.	Issue B Version 1.2	Updated to include details of sections Planning, Design, Development	Process enhancement	Santosh V 09-Apr-25	Jegajith M 09-Apr-25	Sanmangala MS 14-May-25
5	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-08-2025	Santosh V 08-Aug-25	Sanmangala MS 14-08-2025

1. Purpose

The purpose of this document is to define a procedure for New Product Development Lifecycle at Utthunga

2. Scope

The scope of this process extends to all products under development at Utthunga. This methodology can be deployed for Comprehensive end-to-end development encompassing all stages outlined and Selective application of stages based on the defined scope of work.

3. Abbreviations

a. PO – Product owner

b. PM- Project Manager

c. ROI- Return on Investment

d. MVP- Minimal Viable Product

e. RC – Release Candidate

f. SME – Subject Matter Expert

4. Terms and Definitions

a. Product Ideation: refers to the process of generating, developing, and refining ideas for new products or product features.

b. Product Conceptualization: refers to the phase in the product development process where the initial idea or concept for a new product is refined, shaped, and developed into a more concrete and detailed form.

c. Product Design: refers to comprehensive and iterative process of creating and developing a product from its initial concept to its final form.

d. Product Launch: refers to the process of introducing a new product or a significant update to an existing product to the market.

5. Responsibilities

a. Team of Experts (Stakeholders: Users, Customers, Developers, Designers, Marketers, Managers) conduct analysis for idea generation.

b. Product Owner:

- Gathers requirements from end users or SMEs.

- Documents requirements.
- Creates basic design and prototype roadmap.
- Enhances prototype with detailed architecture design.
- Plans budget based on design information.
- Reviews and approves the product before release.
- Creates product user manual and troubleshooting guidelines.
- Takes inputs from Subject Matter Experts.
- Defines warranty terms and conditions.
- Defines maintenance strategy.

c. Development Team:

- Develops actual prototype and conducts preliminary testing.
- Makes iterations based on feasibility study and analysis findings.
- Conducts maintenance activities.

d. Quality Team:

- Conducts work product audits.

e. BU Head:

- Approves the final prototype.

f. Product Manager:

- Conduct analysis for idea generation with help of experts.
- Involves Sales and Marketing teams.
- Sets budget and estimated ROI.
- Maintains the product post-launch.
- Conducts retrospective meetings.
- Reviews and approves the overall product before release.

g. Project Manager:

- Plans roadmap, estimation, and time plan for development.

h. Marketing Team:

- Executes campaigns.

i. Legal Team:

- Reviews and approves warranty terms.

j. Support Team:

- Assesses and resolves warranty claims.

6. RACI

Process	Responsible	Accountable	Consulted	Informed
Product Ideation	Product Manager	BU head	Technical Architects	Senior Management for Investment approval
Product Conceptualization	Product Owner	Product Manager	End users, SMEs for requirements and feedback	Senior Management
Design	Architect, Product owner	Product Manager		BU head, End user
Development and Testing	Project Manager	Product Owner for roadmap planning	Support team for user manual creation	Product Manager
Production	Production Manager/Production supervisor	BU Head	Design team, Product Owner Vendor,	Senior Management
Delivery and Launch	Product Manager, Sales, and Marketing Team	Delivery Manager for the launch plan	Subject Matter Experts for market analysis	Senior Management
Re-design for Cost Optimization	Architect, Product owner	Product Manager	Production Manager/Production supervisor	Senior Management
Post Launch	Product Manager	Project Manager for maintenance and	Team involved in retrospective meetings	Senior Management

		enhancement		
Product Warranty	Product Owner	Product support team	BU head	Customer
Product Upgrade	Product Owner	Product Manager	Production Manager, BU head	Customer
Product Life extension	Product Owner	Product Manager	Production Manager, BU head	Customer
Maintenance and Support	Product owner	Product Manager	BU head	Customer
Product Retirement	Product owner	Product Manager	BU head	Customer

7. SIPOC

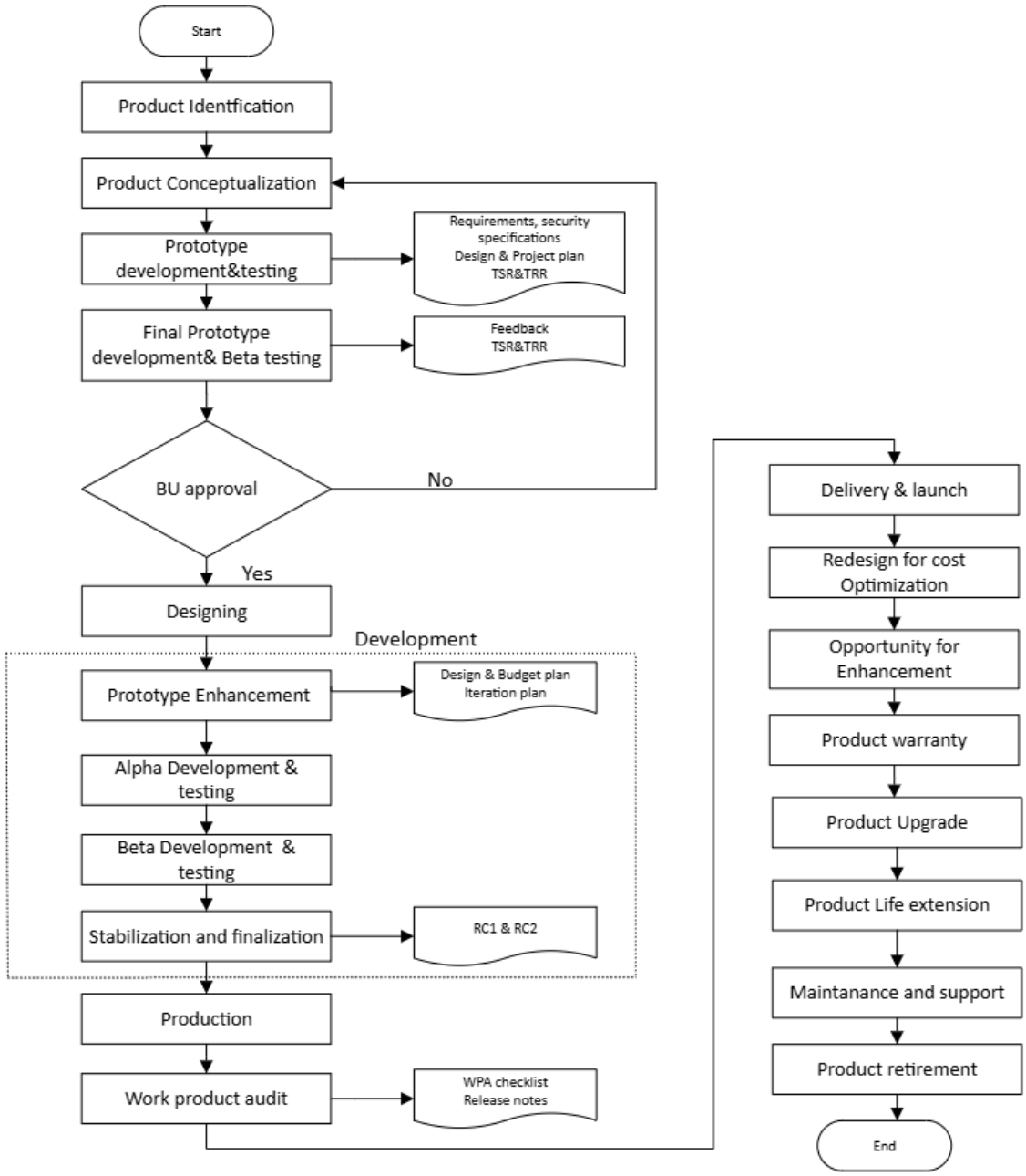
Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Stakeholders, Market Research, Development Team	Ideas, Requirements, Feedback, Budget	Product Ideation	Investment Analysis review and approval by BU head	Approved Ideas	Product Manager
Development Team, End Users, SMEs	quirements, Feedback	Product Conceptualization	Prototype review by product owner	Prototype (Beta release), Final Product Approval	End User
Product Owner, Development Team	Feasibility, Technology Stack, Planning	Design	Detailed Architecture Design review by product owner	Approved Design document	Development team

Development and Testing Team	Design Architecture	Development and Testing	Roadmap Planning, Product Development	Developed Product, Documentation, User Manuals, Troubleshooting	Sales and Marketing team
Design team	Technical documentation, Manufacturing instructions, QC Guidelines, Supply Chain & Logistics	Production	Quality Control	Manufactured product	Product Manager, Sales, and Marketing Team
Product Manager, Sales, and Marketing Team	Market Strategy, Budget, Campaigning	Delivery and Launch plan	Launch Strategy	Released Features, Market Impact	End User
Procurement team, Product Manager	Identified components	Re-design for Cost Optimization	Quality Control	Revised BOM	Production Manager
Product Manager, Development Team	Retrospective Meetings, Enhancement Opportunities	Post Launch	Product Maintenance and Enhancement	Enhanced Product	End User
Product details	Warranty terms & condition, product documentation	Product Warranty	Warranty Coverage Check	Repaired, replaced product	Customers

Product Manager	Upgraded features	Product Upgrade	Quality Control	Upgraded product	Customers
Product Manager	Upgrades	Product Life extension	Quality Control	Updated product	Customers
User	Maintenance contract, product documentation	Maintenance and Support	Quality Control	Enhancements, fixed issues	Customers
Product Owner	Product Performance Evaluation, Market Demand, Profitability	Product Retirement	Product evaluation, Communication to End users and stakeholders	Retired Product and communication to End users	End User

8. Overall Process Flow Chart

a. Product Development Lifecycle



9. Process Description

Product development life cycle management at Utthunga focuses on the process of managing a product from its conception to its launch and beyond. It involves various stages, such as ideation, conceptualization, prototyping, testing, designing, planning, development, delivery, and maintenance.

9.1. Product Ideation

a. Team of experts involving different stakeholders, such as users, customers, developers, designers, marketers, and managers conduct different analysis in a workshop to conclude an idea.

b. Product ideation is done by using various techniques and methods, such as brainstorming, sketching, prototyping, testing and feedback. Refer Design Thinking Guidelines

c. Product ideation should identify customer needs and their segment, market opportunities & size, existing solutions to the problem in the market, potential needs.

d. Selected idea should be evaluated using Investment request Template UTSMOMPTMP09 to check if it is a good Investment for Utthunga and must be reviewed and approved by BU Head.

9.2. Planning

a. Detailed requirements will be captured in product backlog/Requirement specification document as per the updated design.

b. Product roadmap and technical implementation is planned. this includes planning dependencies, third party integration strategy if required and considering all the non-functional requirements needed to build and deliver the product.

c. Roadmap planning shall be created along with project manager which includes preliminary plan (Wireframes & Mockups), prototyping for the sprints and grouping features into the bucket (e.g.: must-have, should-have and nice to have), estimation and time plan for development and resource allocation. Each MVP shall be ready to release to end users once completed.

d. Product shall be developed and tested as per the Utthunga' s Development and Testing process. Refer coding, verification & Validation process.

e. Identify appropriate development methodology

- Waterfall Approach: Sequential process – suitable for well-defined hardware designs.
- Agile Approach: Iterative development – suitable for software and modular hardware designs.

- Hybrid Approach: Combination of both, with agile used for firmware and Waterfall for hardware.

9.3. Product Conceptualization

a. Selected idea shall be conceptualized. The concepts shall be tested and built in two sub steps: 1. Prototyping 2. Testing and Analysis

b. Product owner gathers the requirements by brainstorming or connecting with end users or SMEs. Customer needs through surveys, interviews, focus groups. Identify key pain points and feature requests

c. Gathered requirement shall be documented in requirement specification document or in product backlog based on the type of development methodology used for execution. Waterfall, Agile and hybrid based on requirements.

d. Identify cybersecurity needs and integrate security considerations into the product design. This should be based on scope of activity involved. Following are the possible scenarios

   - Complete design with Utthunga and owns the IP
   - Utthunga does the design and development, but IP is owned by customer. Measure Refer Security Management Guideline UT/IEC/GDL/01

e. Basic design shall be created based on the requirements. Consider the security requirements using SOP – Secure Design UTIIECSOP02. Design shall be reviewed and approved using Design documentation template UTSDDSPTMP01.

f. Prototype roadmap shall be created for implementing the prototype and shall be captured in Utthunga' s project planning template.

g. Prototyping (Proof of Concept): Build early prototypes (low fidelity or technical PoC) to validate critical assumptions.

h. Actual prototype shall be developed and preliminary testing for investigating the performance model is conducted.

i. Test results are captured in TSR and TRR.

j. Feedback from the user is collected to refine the prototype to arrive at the final prototype system.

k. Prototype system must be thoroughly tested for usability, Value, Convenience, addressing customer issues and cost reduction.

l. Tested Final prototype (MVP) is distributed for user experience (Beta release)

m. Based on the Test reports and User feedback, Actual product development approval shall be provided by BU head

9.4. Design

9.4.1. Specification Definition

a. Scope of Application:

- **Intended Use Cases**: Clearly articulate how the product will be used, addressing both primary and secondary functions. For instance, if it's a consumer product, define its primary purpose and potential secondary applications
- **Performance requirements**: Define how well the hardware needs to perform under normal operating conditions.
- **Environmental Specifications:** Define operating conditions (temperature, humidity, exposure to dust or water, etc.).
- **Size and form factors**
- **Materials**
- **Industry Standards**: Identify relevant industry regulations, safety standards, security, regulatory and performance benchmarks that the product must be certified. For example, for medical devices, you might need to adhere to FDA regulations, while for consumer electronics, CE, FCC, UL RoHS, ISO/IEC certifications might be required.
- **Customer Expectations:** Consider customer needs, expectations, and preferences. This includes aesthetics, usability, product features, performance, and reliability. Collect customer feedback through surveys or focus groups, if possible, to guide design.

b. Delivery & Project Timeline: Establish key milestones, phase wise deadlines and resource allocation

- **Key Milestones**: Identify critical points in the project where progress needs to be assessed, or decisions need to be made. For example:
  - Completion of initial design and concept approval.
  - Prototype testing and feedback review.
  - Final design freeze.
  - Initial production run.
  - Final product launch.
- **Phase-Wise Deadlines:** Break the project into phases (e.g., Design Phase, Prototype Phase, Production Phase, Testing Phase) and establish clear deadlines for each. Ensure these deadlines align with the resource availability and project priorities.
- **Resource Allocation:** Determine the resources (human, financial, and material) required for each phase. This includes assigning personnel to specific tasks, procuring raw materials, and budgeting for each phase of development. Ensure resources are optimized to avoid bottlenecks and delays

9.5. Development and Testing

9.5.1. Prototyping

a. Product Owner enhances the Prototype to have a detailed architecture design. This includes defining the feasibility, technology stack, and competence required, identifying dependencies, and planning for development and testing.

b. Architecture design shall be captured in Utthunga' s design template.

c. Budget shall be planned by Product owner based on all the information gathered from the design and preliminary planning for building and launching the actual product.

d. Based on the feasibility study and analysis findings, development team shall make the iterations to improve the design.

e. Improved design shall be reviewed and approved Product manager.

9.5.2. Alpha development

a. Alpha Development: Develop core features and architecture. Limited functionality but demonstrates main workflows.

b. Alpha Release: Internal or limited external release for early feedback. VoC insights are revisited. The output includes Alpha build, internal test plan, early documentation

c. Functional Testing (Internal QA): Verify core functionality works as intended. Output includes Bug reports, functional test results.

d. Initial Compliance Testing: Begin validating against required standards (e.g., IEC, FCC, ISO, etc.) to identify early issues

9.5.3. Development Phase 2: Beta

a. Pre-Beta Enhancements: Address alpha feedback, stabilize features

b. Beta Release (Public/Private): Wider audience than alpha. Used to gather usability and reliability data. Output includes Beta version, release notes, user onboarding material

c. Pre-Compliance Testing: Perform deeper compliance validation (e.g., EMC, cybersecurity pre-assessments.

d. Beta1 Release (Refined Beta): Incorporates fixes from beta testers. Candidate for RC path if stable.

9.5.4. Stabilization & Finalization

a. Release Candidate 1 (RC1): First build that is feature-complete and potentially shippable.

b. Preproduction (Pilot Run): Small-scale production or deployment to simulate mass production or launch. Used for operational testing.

c. Release Candidate 2 (RC2): Includes minor fixes from RC1/preproduction testing. Ideally the final version for production.

9.6. Production

Refer SOP Manufacturing and Assembly UTMFGSOP02, SOP Production process UTMGSOP01 as applicable

9.7. Delivery and Launch

a. Product Manager shall involve Sales and Marketing team for planning the strategy to reach the end users.

b. Product Manager shall set the budget and estimated ROI for Management approval.

c. Marketing team shall start the campaigning upon the budget approval from management.

d. Product Launch plan shall include details on whether to release the new feature to the entire customer base all at once or to conduct a round of beta testing involving a select group of users.

e. Product owner shall take inputs from Subject matter expert to identify the segments and the market to launch the feature/product with maximum impact.

9.8. Re-design for Cost Optimization

a. Evaluate design modifications for reducing cost and improving manufacturability using Value Engineering Guideline UTSDPDLCGL01

- Identify the cost drivers and evaluate for manufacturing challenges and inefficiencies
- Assess material alternatives, explore design simplification and standardization of components
- Create new prototypes with new design and conduct rigorous testing to ensure no compromise on functionality or quality
- Gather feedback from manufacturing and iterate for optimization
- Validate the modifications against the specifications
- Update all design and manufacturing documentation

- Monitor the performance of the modified design in production

9.9. Post Launch

a. Product manager is responsible for maintaining the product post the launch phase.

b. Product owner shall conduct retrospective meetings with the team who worked on the product.

c. Product owner shall see the opportunity to further enhancing the product.

9.10. Product Warranty

a. Product owner defines the warranty terms and conditions, including the warranty duration, coverage, and any limitations or exclusions.

b. Warranty terms and conditions shall be reviewed and approved by legal team.

c. This information is documented and communicated to customers through product documentation and warranty agreements.

d. Customers can report defects or issues and request warranty service using published email id.

e. When a customer makes a warranty claim, the company's support team or service centers assess the issue.

f. If the issue is covered under the warranty, Utthunga provides repair, replacement, or other appropriate solutions.

g. Warranty claims shall be tracked in the template Warranty details UT/EC/SOL/TMP/10 and analyzed to identify patterns and recurring issues and shall be used as an input data for product improvement and quality control.

h. The warranty period specifies how long the manufacturer or provider will support the product. Warranty periods may vary from product to product and can be influenced by local laws and industry standards.

9.11. Product Upgrade

a. Identify the opportunities and need for upgrading the product so that changes add value to customer and align with business goals as per Product upgrade guidelines

9.12. Product Life extension

a. Identify scenarios for where product life requires extension and choose appropriate Life extension strategy as per Product life extension guidelines

9.13. Maintenance and Support

b. Product owner defines the terms and conditions, including the warranty duration, coverage, and any limitations or exclusions.

c. Product maintenance strategy shall be defined to maintain the product as per SOP Lifecycle Services UTSDLCSSOP.

d. Product Documentation shall be made mandatory to plan product maintenance strategy. MARKDOWN Documentation shall include information about how the product/code works, solutions to potential problems.

e. Maintenance of product include various activities such as fixing bugs, adding new features/enhancements, improving performance/performance optimization, porting/migration, re-engineering or updating the software to work with the new hardware or software systems.

9.14. Product Retirement

a. Product Owner shall conduct evaluation of products performance, market demand, profitability, and potential alternatives to decide the product retirement.

b. Communication shall be given to customers, partners & employees about product retirement and timeline.

c. Support and service shall be provided to existing customers for a defined period to ensure smooth transition.

d. Product owner shall capture & document the knowledge and insight gained from product's lifecycle.

e. Product shall be phased out from the market by considering inventory management, contractual obligation and customer transition.

f. Product archival will be carried out as defined in Archival Process.

10. Information Security requirements

a. In the risk management a set of security controls that protect the confidentiality, availability, and integrity of assets from threats and vulnerabilities needs to be identified and mapped by the risk assessors.

11. References

a. SOP- Backup Security and Archival

b. Security Management Guideline UT/IEC/GDL/01

c. SOP – Secure Design UTIIECSOP02

d. Manufacturing and Assembly UTMFGSOP02

e. Production process UTMGSOP01

f. Value Engineering Guideline UTSDPDLCGL01

g. SOP Lifecycle Services UTSDLCSSOP.

12. Metrics

a. Idea Evaluation:

- Number of ideas generated.
- Success rate of ideas selected.

b. Conceptualization and Design:

- Time taken for prototype development.
- Number of design iterations.

c. Development and Testing:

- Refer UTT Development SOP

d. Delivery and Launch:

- Timeliness of launch.
- Customer satisfaction post-launch.

e. Post-Launch:

- Number of enhancements post-launch.
- Customer feedback and satisfaction.

f. Product Warranty:

- Number of warranty claims.

- Resolution time for warranty claims.

g. Maintenance and Support:

- Time and resources spent on maintenance.
- Customer satisfaction with support.

13. Annexures

NA

14. Forms

a. Investment Request Form UTSMOMPTMP09

b. Product Backlog UTSDRSPTMP07

c. Requirement Specification Document UTSDRSPTMP01

d. Project Planning document UTPMPPPTMP04

e. Design documentation template UTSDDSPTMP01

f. Change request form UTPSCMPTMP03

g. Warranty Details UTECSOLTMP10